

ML LAB- 5 Linear Regression - Example Case

CH.SC.U4CSE24118

Ganath Avinash G R

```
import numpy as np # linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
```

Importing Libraries

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
from google.colab import files
```

Reading Dataset Via Kaggle

```
files.upload()
```

[Choose Files](#) salary_Data.csv

salary_Data.csv(text/csv) - 423 bytes, last modified: 5/7/2026 - 100% done

Saving salary_Data.csv to salary_Data (1).csv

{'salary_Data (1).csv':

b'YearsExperience,Salary\n1.1,39343.00\n1.3,46205.00\n1.5,37731.00\n2.0,43525.00\n2.2,39891.00\n2.9,56642.00\n3.0,60150.00\n3.2,54445.00\n3.2,64445.00\n3.7,57189.00\n3

```
data=pd.read_csv("salary_Data.csv")
data.head()
```

	YearsExperience	Salary	
0	1.1	39343.0	
1	1.3	46205.0	
2	1.5	37731.0	
3	2.0	43525.0	
4	2.2	39891.0	

Next steps:

[Generate code with data](#)

[New interactive sheet](#)

Seeing the Dimensions

```
data.shape
```

```
(30, 2)
```

Data Cleaning - Seeing for null values (All values present so 0)

```
data.isnull().sum()
```

```

      0
YearsExperience  0
Salary         0
dtype: int64
```

Assigning Independent Variables & Dependent Variables

```
x=data.iloc[:, :1].values
y=data.iloc[:, 1:2].values
```

Splitting the dataset into 2

```
x_train,x_test,y_train,y_test=train_test_split(
    x,y,
    test_size=0.2,
    random_state=42,
)
```

Fit the model

```
model=LinearRegression()
model.fit(x_train,y_train)
```

```
▼ LinearRegression ⓘ ?
LinearRegression()
```

Testing model

```
y_pred=model.predict(x_test)
y_pred
```

```
array([[115790.21011287],
       [ 71498.27809463],
       [102596.86866063],
       [ 75267.80422384],
       [ 55477.79204548],
       [ 60189.69970699]])
```

Actual vals

```
y_test  
  
array([[112635.],  
       [ 67938.],  
       [113812.],  
       [ 83088.],  
       [ 64445.],  
       [ 57189.]])
```

Almost Ok, but yet there is some error % error in (act-pred) values

Plot for Training set

```
plt.scatter(x_train,y_train,color="red")  
plt.plot(x_train,model.predict(x_train),color="blue")  
plt.title("Salary vs Experience (Training Set)")  
plt.xlabel("Years of Experience")  
plt.ylabel("Salary")  
plt.show()
```



Plot of test set

```
plt.scatter(x_test,y_test,color="red")  
plt.plot(x_test,model.predict(x_test),color="blue")  
plt.title("Salary vs Experience (Test Set)")
```

```
plt.xlabel("Years of Experience")  
plt.ylabel("Salary")  
plt.show()
```

